Weekly Lesson Plan – Week 4 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: August 31, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.406(4) communicate findings and proposed solutions using appropriate tools; §127.406(3) analyze and interpret data to identify features, patterns, and relationships; §127.406(6) implement a design process to develop a multi-system product or solution; §127.406(11) develop, implement, and document repeated trials and tests; §127.406(3) analyze and interpret data; (4) develop evidence-based explanations
ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY AUG 31	TUESDAY SEP 1	WEDNESDAY SEP 2	THURSDAY SEP 3	FRIDAY SEP 4
Students will create a freeCodeCamp account with their school credentials and complete the first two course items.	Students will explain why readable code matters, apply naming conventions, and begin their first guided build.	Students will follow a guided build to produce a working program and identify its input, process, and output.	Students will build a working program from a goal statement without step-by-step instructions and debug it using error messages.	Students will complete course items 1 through 4 and explain how their own program works line by line.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing. Students will…*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.

Assessment of Content Objective: *Instrument used to determine mastery. Ex: Exit Ticket.*

Students will be assessed by written, group, or oral responses throughout the class period.

Set (Bell Work): *A motivating action that engages students in the objective – question, fact, activity. Essential Question:*

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Where is the line between using AI and cheating on your own work?	If AI can already do the entry-level version of a job, should companies still hire people to learn it?	An AI system makes a mistake and a real person gets hurt. Who is responsible - the person who used it, the company that built it, or nobody?	Should an AI have to tell you it is an AI when it talks to you?	Name one decision AI should never be allowed to make on its own. Defend it.

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment. Teacher will… Teams will… Individuals will…*

MONDAY AUG 31	TUESDAY SEP 1	WEDNESDAY SEP 2	THURSDAY SEP 3	FRIDAY SEP 4
Step 1: Bell ringer / essential question. Step 2: Bell work: where is the line between using AI and cheating? (10 min) Step 3: Why we start with code: Fusion licenses are not here yet (5 min) Step 4: Account setup, six steps, rolling by row (12 min) Step 5: Find the course: Full Stack Developer -> JavaScript -> Variables and Strings (5 min) Step 6: Item 1: Introduction to JavaScript (8 min) Step 7: Post the daily freeCodeCamp screenshot (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: should companies still hire people for jobs AI can do? (10 min) Step 3: Code is read more than it is written (8 min) Step 4: The stuck protocol (4 min) Step 5: Item 3: Understanding Code Clarity, then start item 4 (18 min) Step 6: Post the daily freeCodeCamp screenshot (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: an AI hurts someone - who is responsible? (10 min) Step 3: How a Workshop differs from Theory + Input/Process/Output (7 min) Step 4: Item 4: Build a Greeting Bot - type every line (23 min) Step 5: Post the daily freeCodeCamp screenshot (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: should an AI have to tell you it is an AI? (10 min) Step 3: How a Lab works: goal, no steps - plan before you type (5 min) Step 4: Reading errors: line numbers, unexpected token, not defined (5 min) Step 5: Lab: JavaScript Trivia Bot (20 min) Step 6: Post the daily freeCodeCamp screenshot (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: one decision AI should never make on its own (10 min) Step 3: Joining strings: the plus sign and the missing space (5 min) Step 4: Lab: Sentence Maker + finish anything open (18 min) Step 5: Explain your code: point at the input line, point at the output line (7 min) Step 6: Post the daily freeCodeCamp screenshot (5 min)

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state what they accomplished and what worked best.

Study Skills: *Skills implemented by students – note-taking, summarizing, teamwork, etc.*

Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.

SPED & 504 Accommodations / Modifications: *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

ELL Accommodations: *Codes.*

Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).